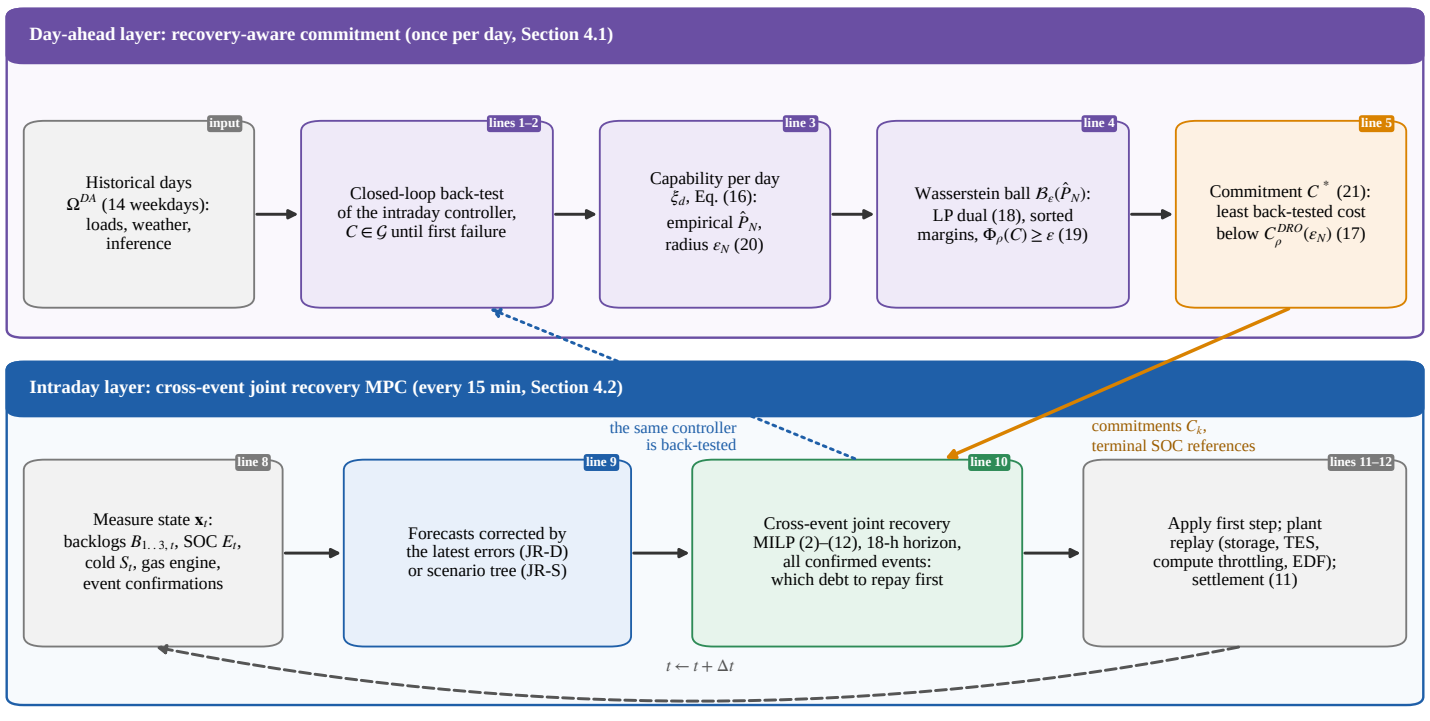
Day-ahead layer: recovery-aware commitment (once per day, Section 4.1)
input
Historical days $\Omega^{DA}$ (14 weekdays): loads, weather, inference
lines 1–2
Closed-loop back-test of the intraday controller, $C \in \mathcal{G}$ until first failure
line 3
Capability per day $\xi_d$, Eq. (16): empirical $\hat{P}_N$, radius $\varepsilon_N$ (20)
line 4
Wasserstein ball $\mathcal{B}_\varepsilon(\hat{P}_N)$: LP dual (18), sorted margins, $\Phi_\rho(C) \geq \varepsilon$ (19)
line 5
Commitment $C^*$ (21): least back-tested cost below $C_\rho^{DRO}(\varepsilon_N)$ (17)
Intraday layer: cross-event joint recovery MPC (every 15 min, Section 4.2)
the same controller is back-tested
commitments $C_k$, terminal SOC references
line 8
Measure state $\mathbf{x}_t$: backlogs $B_{1..3,t}$, SOC $E_t$, cold $S_t$, gas engine, event confirmations
line 9
Forecasts corrected by the latest errors (JR-D) or scenario tree (JR-S)
line 10
Cross-event joint recovery MILP (2)–(12), 18-h horizon, all confirmed events: which debt to repay first
lines 11–12
Apply first step; plant replay (storage, TES, compute throttling, EDF); settlement (11)
$t \leftarrow t + \Delta t$